Digital Logic Design

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Digital Logic Design deals with the analysis and design of Digital systems.

Digital system means that performs operations based on the principles of digital techniques.

Calculator is an example of digital system.

Digital signal can have two logic levels called as LOW level & HIGH level.

They are also represented as logic '0' or logic '1'.

<u>Advantages of Digital systems</u>:-

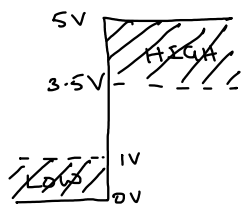
1. Digital system design is easier when compared to analog systems because digital systems operate on one of the two states called as ON & OFF.
2. Digital techniques require Boolean algebra which is very simple and easy to understand.
3. Storage of information is easier
4. Large number of ICs are available for performing various operations.

   They are
   - Reliable
   - Accurate
   - Small in size
   - Speed of operation is very high.
5. More immune to noise.
6. Display of data & other information is very accurate and convenient.

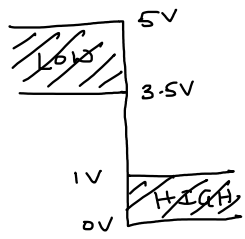
Because of these advantages, most of the latest electronic systems are digital in nature.

In general digital signals can be represented in two ways.

1. Positive logic system
2. Negative logic system.



a) Positive Logic



b) Negative Logic

In Positive logic system, any voltage between 3.5V to 5V is considered as logic HIGH & any voltage between 0 to 1V is considered as logic LOW level.

In positive logic system, any voltage between 3.5V to 5V is [illegible]
Logic HIGH & any voltage between 0 to 1V is considered as logic LOW level.
In positive logic system, high voltage is considered as logic '1' & low voltage is considered as logic '0'.
where as in negative logic system, any voltage between 3.5 to 5V is considered as logic LOW & any voltage between 0 to 1V is considered as Logic HIGH level.
ie In negative logic system, high voltage is considered as logic '0' & low voltage is considered as logic '1'.

Number systems :-

Number systems are classified as

1. Decimal number system
2. Binary number system
3. octal number system
4. Hexadecimal number system.

Number system	Base or Radix	Digits used
Decimal	10	0, 1, 2, 3, 4, 5, 6, 7, 8, 9
Binary	2	0, 1
octal	8	0, 1, 2, 3, 4, 5, 6, 7
Hexadecimal	16	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F

In general, the number has two parts integers and fractional parts separated by a radix point i.e

$$(N)_r = \underbrace{d_{n-1}\, d_{n-2} \cdots d_2\, d_1\, d_0}_{\text{Integer part}} \cdot \underbrace{d_{-1}\, d_{-2}\, d_{-3} \cdots d_{-m}}_{\text{fractional part}}$$

Radix Point

where

N is a number
r is the base or radix of number system.
n is the number of digits in the integer part
m is the number of digits in the fractional part
d_{n-1} is called Most significant Digit (MSD)
d_{-m} (LSD)

d_{n-1} is called Most significant Digit (MSD)

d_{-m} is called as Least significant Digit (LSD)

$$N = d_{n-1} r^{n-1} + d_{n-2} r^{n-2} + \cdots + d_2 r^2 + d_1 r^1 + d_0 r^0 + d_{-1} r^{-1} + d_{-2} r^{-2} + \cdots + d_{-m} \cdot r^{-m}$$

For ex :- $10^3\ 10^2\ 10^1\ 10^0\ 10^{-1}\ 10^{-2}\ 10^{-3}$ ← weights assigned to the positions

i) $(9763.245)_{10}$

$$= 9\times10^3 + 7\times10^2 + 6\times10^1 + 3\times10^0 + 2\times10^{-1} + 4\times10^{-2} + 5\times10^{-3}$$

$$= 9763.245$$

ii) $2^3\ 2^2\ 2^1\ 2^0\ 2^{-1}\ 2^{-2}$

$$(1101.11)_2 = 1\times2^3 + 1\times2^2 + 0\times2^1 + 1\times2^0 + 1\times2^{-1} + 1\times2^{-2}$$

$$= 8 + 4 + 0 + 1 + \tfrac{1}{2} + \tfrac{1}{4}$$

$$= (13.75)_{10}$$

iii) $5^3\ 5^2\ 5^1\ 5^0$

$$(2314)_5 = 2\times5^3 + 3\times5^2 + 1\times5^1 + 4\times5^0$$

$$= 250 + 75 + 5 + 4$$

$$= (334)_{10}$$

iv) $(1201.12)_3 = (?)_{10}$

$$= 1\times3^3 + 2\times3^2 + 0\times3^1 + 1\times3^0 + 1\times3^{-1} + 2\times3^{-2}$$

$$= 27 + 18 + 0 + 1 + \tfrac{1}{3} + \tfrac{2}{9}$$

$$= (46.55)_{10}$$

<u>Binary Number system</u> :- A number system with base or radix '2' is called as Binary number system.

The digits that are used in this system are 0 & 1. These digits are called as Binary digits or Bits.

Group of 4 bits is called as Nibble.

Group of 8 bits is called as Byte.

Decimal number	Binary Representation
0	0
1	1
2	01
3	11
4	100
	101

3	11
4	100
5	101
6	110
7	111
8	1000
9	1001
10	1010
11	1011
12	1100
13	1101
14	1110
15	1111
16	10000

If 'n' is the number of bits in a binary, then the max. decimal number that can be represented with 'n' bits is $2^n - 1$.

$$\underbrace{\overset{2^{n-1}}{d_{n-1}}\ \overset{2^{n-2}}{d_{n-2}}\ \cdots\ \overset{2^2}{d_2}\ \overset{2^1}{d_1}\ \overset{2^0}{d_0}}_{\text{Integer portion}}\ \underset{\text{Binary Point}}{\cdot}\ \underbrace{\overset{2^{-1}}{d_{-1}}\ \overset{2^{-2}}{d_{-2}}\ \cdots\ \overset{2^{-m}}{d_{-m}}}_{\text{Fractional portion}}$$

$$\therefore N = d_{n-1} \times 2^{n-1} + d_{n-2} \times 2^{n-2} + \cdots + d_2 \times 2^2 + d_1 \times 2^1 + d_0 \times 2^0 + d_{-1} \times 2^{-1} + d_{-2} \times 2^{-2} + \cdots + d_{-m} \times 2^{-m}$$

<u>Binary to Decimal Conversion</u>:- Binary number can be converted into its decimal number using the weights assigned to each bit position.

1) $(1101111)_2 = (?)_{10}$

$$(\overset{2^6}{1}\overset{2^5}{1}\overset{2^4}{0}\overset{2^3}{1}\overset{2^2}{1}\overset{2^1}{1}\overset{2^0}{1})_2 = 1\times2^6 + 1\times2^5 + 0\times2^4 + 1\times2^3 + 1\times2^2 + 1\times2^1 + 1\times2^0$$

$$= (111)_{10}$$

2) $(10010.1011)_2 = (?)_{10}$

$$(\overset{2^4}{1}\overset{2^3}{0}\overset{2^2}{0}\overset{2^1}{1}\overset{2^0}{0}.\overset{2^{-1}}{1}\overset{2^{-2}}{0}\overset{2^{-3}}{1}\overset{2^{-4}}{1})_2 = 1\times2^4 + 0\times2^3 + 0\times2^2 + 1\times2^1 + 0\times2^0 + 1\times2^{-1} + 0\times2^{-2} + 1\times2^{-3} + 1\times2^{-4}$$

$$= 16 + 0 + 0 + 2 + 0 + \frac{1}{2} + 0 + \frac{1}{8} + \frac{1}{16}$$